

Jacob Kang

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Research Interest

- Computational Neuroscience
- Brain Imaging
- Machine Learning
- Nonstationary Dynamics

Education

University of Maryland, College Park, Ph.D. in Bioengineering Aug 2026 –

- Doctoral Advisor: Prof. Nan Xu
- Relevant Coursework: Network Neuroscience and Dynamic Analysis, Machine Learning and Statistical Signal Processing, Random Processes in Communication and Control

University of Maryland, College Park, B.S. in Computer Science and Mathematics: Statistics Jan 2024 – May 2026

- GPA: 3.91/4.00; Graduated with **Cum Laude**
- Academic Mentor: Prof. James A. Yorke
- Relevant Coursework: Computer Vision, Bioinformatics Algorithm, Numerical Analysis, Statistical Computing with SAS, Sampling Theory, Application of Linear Algebra, Advanced Data Structure, Advanced Calculus, Differential Equations

Pennsylvania State University, University Park, Studied Computational Data Science Aug 2022 – Dec 2023

- GPA: 3.91/4.00
- Relevant Coursework: Data Science through Statistical Reasoning, Data Management for Data Science, Programming Models for Big Data, Probability Theory

Publications

† These authors have contributed equally to this work and share first authorship

Hoheisel L. †, **Kang J.** †, Sathiyamoorthy A., Mandino F., Bernat E., Lake E., Xu N. A conserved neural–BOLD brain state reveals glia-associated neurovascular coupling. In preparation for *Nature Neuroscience*, 2026. (Co-first author)

Jian J., Multezem N., Li B., **Kang J.**, Xu N. Beyond Gaussian assumptions: distribution-aware channel capacity for brain effective connectivity. Submitted to *NeurIPS*, 2026.

Kang H. †, **Kang J.** †, Zeki M., Seo J. Clustered pattern projection for EEG dementia classification: evaluating the generalizability of disorder patterns. *Frontiers in Neuroscience*, accepted, 2026. (Co-first author)

Seo J., Kang H., **Kang J.**, Zreikat A. I. Dynamic mode decomposition–based clustered pattern projection for reliable Alzheimer’s disease detection from EEG. *Diagnostics*, 2026. doi:10.3390/diagnostics16040530

Kang J., Kang H., Seo J. CNN-based framework for Alzheimer’s disease detection from EEG via dynamic mode decomposition. *Frontiers in Neuroinformatics*, 2025. doi:10.3389/fninf.2025.1706099 (First author)

Presentations

Kang J., Hoheisel L., Sathiyamoorthy A., Mandino F., Bernat E., Lake E., Xu N. A reproducible neural–BOLD state motif links brain-state dynamics across modalities and species. *BRAIN Initiative*, accepted, 2026.

Hoheisel L., **Kang J.**, Sathiyamoorthy A., Mandino F., Bernat E., Lake E., Xu N. Selective neural–BOLD state correspondence reveals a conserved brain-state motif across modalities and species. *Society for Neuroscience*, submitted, 2026.

Hoheisel L., **Kang J.**, Sathiyamoorthy A., Bernat E., Xu N. Brain states in simultaneous EEG–fMRI. *Organization for Human Brain Mapping*, 2026.

Kang J., Hoheisel L., Xu N. Temporal coupling of neural and BOLD dynamic states from simultaneous resting-state EEG–fMRI. JHU & UMD Mid-Atlantic Research Day / UMD Undergraduate Research Day, 2026. Poster | Abstract

Kang J., Zhang K., Weitz J. Modeling the dynamics of breakthrough infections and vaccination stagnation. UMD Undergraduate Research Day, 2026. Poster

Research Experiences

Undergraduate Researcher, INSPIRE Lab
University of Maryland, College Park
Mentored by Prof. Nan Xu, Linnea Hoheisel (Postdoctoral Researcher)

- Found transient cross-frequency neural-state co-expression pattern that is frequently co-occurring with a single BOLD state that exhibits prevalent negative engagement of the default mode and frontoparietal network around onset of the co-expression
- Validated with two independent simultaneous EEG-fMRI data in humans (n = 22 and 12; TR = 2 s and 2.52 s) and calcium fMRI data in mice and observed reproducible pattern across modalities and species
- Developed Hidden Markov Model-based brain state analysis framework that examines recurring brain states from neural, cellular, hemodynamic signals and support future biomarker studies in disorders involving frontoparietal dysfunction (e.g. ADHD), enabling more accessible investigation of large-scale brain dynamics

Nov 2025 – May 2026

REU Student & Undergraduate Researcher, Weitz Group
University of Maryland, College Park
Mentored by Prof. Joshua Weitz, Kejia Zhang (PhD Student)

- Developed SIRV epidemic model implementing both deterministic and stochastic simulations to evaluate vaccination strategies across parameter space of R_0 and vaccination rate (v)
- Identified critical vaccination thresholds and optimal coverage strategies that maximize epidemic control while minimizing resource allocation; presented findings at UMD Summer REU Poster Session
- Extended model to incorporate vaccine failure rates, revealing a counterintuitive threshold: below critical failure rates, peak breakthrough infections occur at intermediate (not maximal) vaccination coverage

Jun 2025 – Jun 2026
github.com/mdsnpr/Epidemiology_UMD

Lead Researcher, Independent Research
Mentored by Dr. Jonghyeon Seo

- Developed Alzheimer's classification framework using EEG data from 63 subjects (36 patients, 27 controls), combining Dynamic Mode Decomposition with Sequential CNN to achieve 74.23% accuracy in Leave-One-Subject-Out validation, outperforming recent baseline (62.50%) and FFT-based methods (70.56%)
- Adopted stacked shortened window (2 seconds * 12) approach when forming input to 3D CNN to preserve nonstationary dynamics, especially emphasized in eyes open EEG data
- Identified delta (δ) frequency band as a confounding factor in Alzheimer's classification; excluding δ resulted in statistically significant improvement in accuracy by 1.27% ($p < 0.05$, paired t-test)
- Led further project that incorporates clustered pattern projection to improve classification reliability and resulted in two first author and one co-author publication

May 2025 – Jul 2026
github.com/mdsnpr/AlzDetect_EEG_DMD

Work and Teaching Experiences

Supervisor, Nebula Research Data Fellowship, University of Maryland
Faculty Assistant, INSPIRE Lab, University of Maryland

- Conducting computational neuroscience research and in-lab IT support for INSPIRE LAB under Prof. Nan Xu

Sep 2026 –
Jun 2026 –

Grader, MATH 246: Differential Equations, University of Maryland
Learning Assistant, DS 200: Introduction to Data Science, Pennsylvania State University

Sep 2025 – Dec 2025
Aug 2023 – Dec 2023

Advising and Mentoring Experiences

Undergraduates

- Gracelyn Arunachalam, Computer Engineering, University of Maryland
 - Nebula Research Data Fellow
- Vishal Senthilkumar, Computer Science, University of Maryland
 - Assisting quasi-periodic spatiotemporal pattern-based computational neuroscience research using fMRI from infants with prenatal opioid exposure.

Sep 2026 –
May 2026 –

Awards and Honors

CMNS Scholarship Funds (\$3000), University of Maryland	2025
President’s Transfer Scholarship (\$10000), University of Maryland	2024
President Walker Award, Pennsylvania State University	2023
Top Prize in Big Data Utilized Idea Competition, Seoul Metropolitan Office of Education	2021
Medal and Certificate in Euclid Mathematics Competition, University of Waterloo	2021

Skills

Programming: MATLAB, Python, R, SAS

Computational Methods: Hidden Markov Models, Dynamic Mode Decomposition, Machine Learning, Time-Series Analysis, Mathematical and Statistical Modeling

References

James A. Yorke Department of Mathematics and Physics, University of Maryland, College Park <i>Academic Mentor</i>	yorke@umd.edu
Nan Xu Fischell Department of Bioengineering, University of Maryland, College Park <i>Doctoral Advisor</i>	nanxu@umd.edu inspirelab.umd.edu
Joshua S. Weitz Department of Biology, University of Maryland, College Park <i>Undergraduate Research Advisor</i>	jsweitz@umd.edu weitzgroup.umd.edu
Lawrence C. Washington Department of Mathematics, University of Maryland, College Park <i>Coordinator of REU program in Infectious Disease Dynamics</i>	lcw@umd.edu